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Submission: *Assignment_1_Linux_Basics*

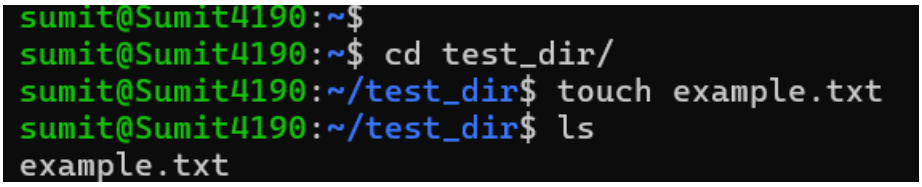
1. Creating and Renaming Files/Directories

1. Create a directory named test_dir using mkdir.



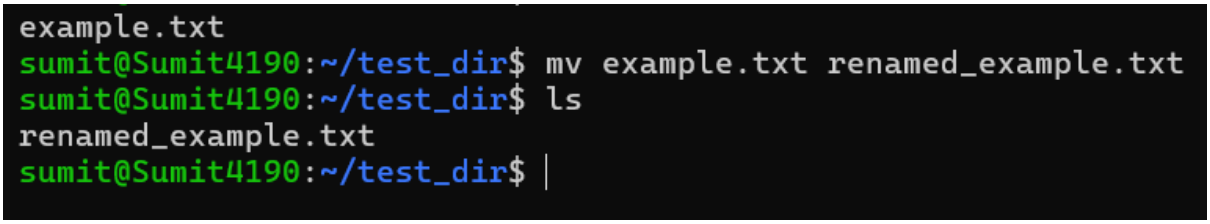
   Here, I have created a test_dir using mkdir in my home dir.

2. Inside test_dir, create an empty file called example.txt.



   Touch cmd is used to create a empty file, although there are other alternative such as nano example.txt – it will create or use the existing file (if file already exist) and open in nano editor and also we can use the vim example.txt.

3. Rename example.txt to renamed_example.txt using mv.



   mv cmd uses the source and destination if we provide the source and destination same it will create a new fill with new name with same content (without changing any content of the file) if we provide the another destination other than source it will create a copy of a file.

2. Viewing File Contents

4. Use `cat` to display the contents of /etc/passwd.

   Cat is used to display the content of the file. If we have to read a file without changing any thing we use the cat cmd.

```
sumit@Sumit4190:~/test_dir$
sumit@Sumit4190:~/test_dir$ cat /etc/passwd
root:x:0:0:root:/root:/bin/bash
daemon:x:1:1:daemon:/usr/sbin:/usr/sbin/nologin
bin:x:2:2:bin:/bin:/usr/sbin/nologin
sys:x:3:3:sys:/dev:/usr/sbin/nologin
sync:x:4:65534:sync:/bin:/bin/sync
games:x:5:60:games:/usr/games:/usr/sbin/nologin
man:x:6:12:man:/var/cache/man:/usr/sbin/nologin
lp:x:7:7:lp:/var/spool/lpd:/usr/sbin/nologin
mail:x:8:8:mail:/var/mail:/usr/sbin/nologin
news:x:9:9:news:/var/spool/news:/usr/sbin/nologin
uucp:x:10:10:uucp:/var/spool/uucp:/usr/sbin/nologin
proxy:x:13:13:proxy:/bin:/usr/sbin/nologin
www-data:x:33:33:www-data:/var/www:/usr/sbin/nologin
backup:x:34:34:backup:/var/backups:/usr/sbin/nologin
list:x:38:38:Mailing List Manager:/var/list:/usr/sbin/nologin
irc:x:39:39:ircd:/run/ircd:/usr/sbin/nologin
_apt:x:42:65534::/nonexistent:/usr/sbin/nologin
nobody:x:65534:65534:nobody:/nonexistent:/usr/sbin/nologin
systemd-network:x:998:998:systemd Network Management:/:/usr/sbin/nologin
systemd-timesync:x:996:996:systemd Time Synchronization:/:/usr/sbin/nologin
dhcpcd:x:100:65534:DHCP Client Daemon,,,:/usr/lib/dhcpcd:/bin/false
messagebus:x:101:101::/nonexistent:/usr/sbin/nologin
syslog:x:102:102::/nonexistent:/usr/sbin/nologin
systemd-resolve:x:991:991:systemd Resolver:/:/usr/sbin/nologin
uuidd:x:103:103::/run/uuidd:/usr/sbin/nologin
landscape:x:104:105::/var/lib/landscape:/usr/sbin/nologin
polkitd:x:990:990:User for polkitd:/:/usr/sbin/nologin
sumit:x:1000:1000:,,,:/home/sumit:/bin/bash
patil:x:1002:1002:Sumit Patil,999,+91 9876543210,+91 1234567890:/home/patil:/bin/bash
sam:x:1003:1003:Sam,,,:/home/sam:/bin/bash
sshd:x:105:65534::/run/sshd:/usr/sbin/nologin
sumit@Sumit4190:~/test_dir$
```

5. Display only the first 5 lines of /etc/passwd using `head`.

```
sumit@Sumit4190:~/test_dir$ head -5 /etc/passwd
root:x:0:0:root:/root:/bin/bash
daemon:x:1:1:daemon:/usr/sbin:/usr/sbin/nologin
bin:x:2:2:bin:/bin:/usr/sbin/nologin
sys:x:3:3:sys:/dev:/usr/sbin/nologin
sync:x:4:65534:sync:/bin:/bin/sync
sumit@Sumit4190:~/test_dir$
```

Head takes the numerical argument which describes how much top line of the file we want to display. If we don't specify the number, it uses default - 10.

6. Display only the last 5 lines of /etc/passwd using `tail`.

```
sumit@Sumit4190:~/test_dir$
sumit@Sumit4190:~/test_dir$ tail -5 /etc/passwd
polkitd:x:990:990:User for polkitd:/:/usr/sbin/nologin
sumit:x:1000:1000:,,,:/home/sumit:/bin/bash
patil:x:1002:1002:Sumit Patil,999,+91 9876543210,+91 1234567890:/home/patil:/bin/bash
sam:x:1003:1003:Sam,,,:/home/sam:/bin/bash
sshd:x:105:65534::/run/sshd:/usr/sbin/nologin
sumit@Sumit4190:~/test_dir$
```

Tail takes the numerical argument which describes how much last line of the file we want to display. If we don't specify the number, it uses default - 10. We can use tail -f to add new upcoming lines of the files on display.

3. Searching for Patterns

7. Use `grep` to find all lines containing the word "root" in /etc/passwd.

Grep cmd is used to filter out the output of the file or cmd . it gives the output of the lines which matches with the condition. We can use the wildcards to search for the similar patterns.

```
sumit@Sumit4190:~/test_dir$ grep root /etc/passwd
root:x:0:0:root:/root:/bin/bash
sumit@Sumit4190:~/test_dir$
```

4. Zipping and Unzipping

8. Compress the test_dir directory into a file named test_dir.zip using `zip`.

```
sumit@Sumit4190:~$ zip -r test_dir.zip test_dir
  adding: test_dir/ (stored 0%)
  adding: test_dir/renamed_zip/ (stored 0%)
  adding: test_dir/renamed_zip/unzipped_dir/ (stored 0%)
  adding: test_dir/renamed_zip/unzipped_dir/test_dir/ (stored 0%)
  adding: test_dir/renamed_zip/unzipped_dir/test_dir/renamed_example.txt.gz (deflated 13%)
  adding: test_dir/secure.txt
zip warning: Permission denied
        zip warning: could not open for reading: test_dir/secure.txt
  adding: test_dir/renamed_example.txt (stored 0%)

zip warning: Not all files were readable
  files/entries read:  6 (40 bytes)  skipped:  1 (0 bytes)
sumit@Sumit4190:~$
```

Zip uses the -r option to select all files of the folder without the -r option zip only select the empty files.

9. Unzip test_dir.zip into a new directory named unzipped_dir.

```
sumit@Sumit4190:~$ unzip test_dir.zip -d /home/sumit/test_dir/renamed_zip/unzipped_dir
Archive:  test_dir.zip
   creating: /home/sumit/test_dir/renamed_zip/unzipped_dir/test_dir/
  inflating: /home/sumit/test_dir/renamed_zip/unzipped_dir/test_dir/renamed_example.txt.gz
sumit@Sumit4190:~$ D
```

Unzip use the -d option to unzip an zip file at a specific location if we want to unzip at current folder we don't have to use the -d we just name the file.

5. Downloading Files

10. Use wget to download a file from a URL (e.g., https://example.com/sample.txt).

```
sumit@Sumit4190:~$ cd test_dir/
sumit@Sumit4190:~/test_dir$ wget https://example.com/sample.txt
--2026-10-28 23:00:15--  https://example.com/sample.txt
Resolving example.com (example.com)... 172.66.147.243, 104.20.23.154, 2606:4700:8d75:72db:f2a7:4ba:ef6b:ff98
Connecting to example.com (example.com)|172.66.147.243|:443... connected.
HTTP request sent, awaiting response... 404 Not Found
2026-10-28 23:00:16 ERROR 404: Not Found.

sumit@Sumit4190:~/test_dir$
```

Wget is resource downloading cmd which downloads the resource from the given url there is a another alternative for this cmd curl it used when we don't want to download the resource instead print it on the console.

6. Changing Permissions

11. Create a file named secure.txt and change its permissions to read-only for everyone using chmod.

```
renamed_example.txt  renamed_zip  secure.txt
sumit@Sumit4190:~/Linux_Basics_SumitPatil/test_dir$ chmod 444 secure.txt
sumit@Sumit4190:~/Linux_Basics_SumitPatil/test_dir$ ls -l
total 4
-rw-r--r-- 1 sumit sumit    0 Oct 28 22:45 renamed_example.txt
drwxr-xr-x 3 sumit sumit 4096 Oct 28 23:08 renamed_zip
-r--r--r-- 1 sumit sumit    0 Oct 28 23:01 secure.txt
sumit@Sumit4190:~/Linux_Basics_SumitPatil/test_dir$
```

To change the permission of a file to read-only for every one we use the 4 which gives the read permission for the user , groups and other.

7. Working with Environment Variables

12. Use export to set a new environment variable called MY_VAR with the value "Hello, Linux!".

```
sumit@Sumit4190:~/test_dir$ export MY_VAR="Hello, Linux!"
sumit@Sumit4190:~/test_dir$ $MY_VAR
Hello,: command not found
sumit@Sumit4190:~/test_dir$ echo $MY_VAR
Hello, Linux!
sumit@Sumit4190:~/test_dir$
```

When you define a normal variable in a terminal, it is only available within that specific shell session. Using export ensures that any programs, scripts, or sub-processes you start from that terminal can also see and use the variable.